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28 **Supplemental eTable 1. Search strategy for each electronic database.**

#1	Search terms	No. of records returned
1	hypertensive disorders during pregnancy	7147
2	hypertensive disorders in pregnancy	7147
3	hypertensive disorders of pregnancy	7147
4	pregnancy-induced hypertension	2672
5	eclampsia	1665
6	pre-eclampsia	8158
7	preeclampsia	10799
8	preeclamptic pregnancy	464
9	chronic hypertension	3270
10	gestational hypertension	2538
11	chronic hypertension with superimposed preeclampsia	200
12	postpartum hypertension	988
13	gravid hypertension	65
14	maternal hypertensive disorders	50
15	(#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14)	18801
16	asthma	53601
17	wheeze	9817
18	wheezing	9817
19	shortness of breath	5823
20	bronchial spasm	1749
21	bronchospasm	1749
22	bronchoconstriction	3413
23	bronchoconstrict	1505
24	bronchial hyperreactivity	5976
25	bronchial hyperresponsiveness	2321
26	reactive airway disease	280
27	obstructive lung disease	4734
28	airflow limitation	280
29	chronic obstructive respiratory disorder	7
30	chronic obstructive pulmonary disease	5110
31	(#16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30)	154781
32	(#15 AND #31)	1729
33	limit #32 to English language	1630
34	limit #33 to humans	1456

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Supplemental eTable 2. Newcastle-Ottawa Scale (NOS) for cohort studies.

Study	Item and scores								Scores and quality
	Representiveness of the exposed cohort（1）	Selection of the non-exposed cohort（1）	Ascertainment of exposure（1）	Demonstration that the outcome of interest was not present at the start of the study（1）	Comparability of cohorts based on the design or analysis（2）	Assessment of outcome（1）	Was follow up long enough for outcomes to occur（1）	Adequacy of follow up of cohorts（1）	
Mirzakhani et al (2019)	0	1	1	1	2	0	0	1	6 (M)
Wilmink et al (2018)	1	1	1	1	1	0	1	1	7 (H)
Stokholm et al (2017)	1	1	1	1	2	1	1	1	9 (H)
Shaheen et al (2016)	1	1	1	1	2	0	1	1	8 (H)
Magnus et al (2016)	1	1	1	1	2	1	1	1	9 (H)
Liu et al (2015)	1	1	1	1	2	1	1	1	9 (H)
Byberg et al (2014)	0	1	1	1	2	0	1	1	7(H)
Aspberg et al (2010)	1	1	1	1	0	1	1	0	6 (M)
Rusconi et al (2007)	1	1	1	1	1	0	1	1	7(H)
Nafstad et al (2003)	1	1	1	1	0	1	1	1	7(H)

Note: The quality of the studies was divided into the following 3 categories: high quality (scores 7–9), moderate quality (scores 4–6) and low quality (scores 0–3).

Abbreviations: M, moderate quality; H, high quality.

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies.

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Mirzakhani et al (2019)	adjusted RR [a]: asthma: 1.62 (1.02–2.57)	/	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
	adjusted HR [b]: asthma: 2.68 (1.30–5.61) asthma and recurrent wheeze: 4.73 (2.20–10.07)	yes	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
Wilmink et al (2018)	adjusted OR: current asthma [c]: 0.99 (0.57, 1.73) (GH); 0.80 (0.32, 2.01) (PE/HELLP) current wheezing [d]: 0.95 (0.63, 1.42) (GH); 0.70 (0.36, 1.39) (PE/HELLP)	/	yes	pre-pregnancy BMI	maternal age, maternal ethnicity, educational level, nulliparity, folic acid use	/	sex

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Stokholm et al (2017)	adjusted IRR: Registry-based cohort: 1.09 (1.05, 1.12)	/	/	/	mode of delivery, parity, multiple births, season of birth, maternal age, and mother's disease [e]	birth weight, gestational age	sex, age and calendar year, and season of birth
	adjusted OR [f]: $COPSAC_{2000}$ Birth Cohort: 1.13 (0.31, 4.13)	yes	/	/	type of delivery	/	/
Shaheen et al (2016)	adjusted OR: 1.20 (0.78, 1.83) (preeclampsia) 1.04 (0.87, 1.24) (GH) 1.33 (1.00, 1.78) (HbP)	yes	/	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex
	adjusted OR: 1.23 (0.80, 1.88) (preeclampsia) 1.05 (0.88, 1.25) (GH) 1.34 (1.00, 1.79) (HbP)	yes	yes	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex

52 **Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).**

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Magnus et al (2016)	adjusted RR: registry-based study: 1.31 (1.22, 1.41)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: registry-based study: 1.29 (1.19–1.39)	yes	yes	/	maternal age, parity, education	/	sex
	adjusted OR: sibling pair analysis: 1.07 (0.87, 1.33)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: MoBa study: 1.19 (0.99–1.44)	yes	yes	pre-pregnancy BMI	maternal age, parity, education	/	sex
Liu et al (2015)	adjusted IRR: nested case-control study: 1.19 (1.15, 1.24)	yes	yes	/	maternal age, maternal social status at birth, parity, and maternal country of origin	/	/
	adjusted IRR: case-sibling study: 1.06 (0.98, 1.14)	/	yes	/	maternal age, maternal social status at birth, and parity	/	sex, calendar year of birth

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Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Byberg et al (2014)	adjusted OR: FU1, asthma: 0.87 (0.38, 1.97) (mild/moderate preeclampsia) 0.72 (0.19, 2.77) (severe preeclampsia)	yes	yes	maternal BMI	cesarean section, maternal age	[g]	sex, being firstborn
	adjusted OR: FU2, current asthma [h]: 1.11 (0.45, 2.73) (mild/moderate preeclampsia) 0.69 (0.15, 3.18) (severe preeclampsia)	yes	yes	maternal BMI	cesarean section, maternal age	[g]	sex, being firstborn
Aspberg et al (2010)	adjusted OR: 1.24 (1.19–1.29)	/	yes	/	maternal age, parity, and involuntary childlessness	/	year of birth
Rusconi et al (2007)	adjusted OR: transient early wheezing [i]: 1.47 (1.14–1.89) persistent wheezing [j]: 1.71 (1.25–2.36) late-onset wheezing [k]: 1.45 (1.06–1.48)	/	yes	/	study center, parental atopy, maternal age, socioeconomic status, season of the questionnaire's completion, and person who completed the questionnaire	/	sex, low birth weight, siblings

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Nafstad et al[l] (2003)	adjusted OR: 1.08 (0.97–1.19) (total)	/	/	/	hyperemesis, maternal age, maternal education, plurality, single parent family, death or emigration during the study period, maternal atopy, diabetes mellitus, lower respiratory tract infections, and other maternal infections during pregnancy, other pregnancy complications [m]	/	sex, birth order, year of birth, geographical district of birth

Note:

[a] Mothers with preeclampsia compared with mothers without preeclampsia;

[b] Mothers with asthma with preeclampsia compared with mothers without asthma without preeclampsia;

[c] Current asthma was defined as previous diagnosis of asthma by physician with either current wheezing or the use of inhaled medication in the past 12 months;

[d] Current wheezing, physician diagnosed asthma and use of inhaled medication (bronchodilators or corticosteroids) in the past 12 months was assessed by a parental questionnaire at the age of 10 years;

[e] Mother's disease, not mentioned;

[f] Mothers with asthma with preeclampsia compared with mothers with asthma without preeclampsia (all mothers were diagnosed with asthma in the whole cohort);

[g] All children were born full term;

[h] Current asthma at FU2 was defined as asthma ever in addition to asthma symptoms or the use of asthma medication during the last 12 months;

[i] Transient early wheezers: those who had at least one lower respiratory tract illness with wheezing (bronchiolitis, asthmatic bronchitis, asthma or wheezing) in the first 2 yr of life and no wheezing in the last 12 mo;

[j] Persistent wheezers: those who had at least one lower respiratory tract illness with wheezing in the first 2 yr of life and wheezing in the last 12 mo;

[k] Late-onset wheezers: those with no lower respiratory tract illness with wheezing in the first 2 yr of life but who had wheezing in the last 12 mo (late-onset wheezers).

[l] Only persons with severe asthma are registered in NIAR;

[m] Pregnancy complications, including hemorrhage, anemia, cervix insufficiency, placenta dysfunction, rhesus immunization, and preterm contractions.

Abbreviations: HDP, hypertension disorders in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; VDRRT, Vitamin D Antenatal Asthma Reduction Trial; ISSHP, International Society for the Study of Hypertension in Pregnancy; ACOG, American College of Obstetricians and Gynaecologists; $COPSAC_{2000}$, Copenhagen Prospective Studies on Asthma in $Childhood_{2000;}$ ALSPAC, Avon Longitudinal Study of Parents and Children; DMBR, Danish Medical Birth Registry; ISAAC, International Study of Asthma and Allergies in Childhood; SIDRIA, Italian Studies of Respiratory Disorders in Childhood and the Environment; MBRN, Medical Birth Registry of Norway; NIAR, National Insurance Administration Register.

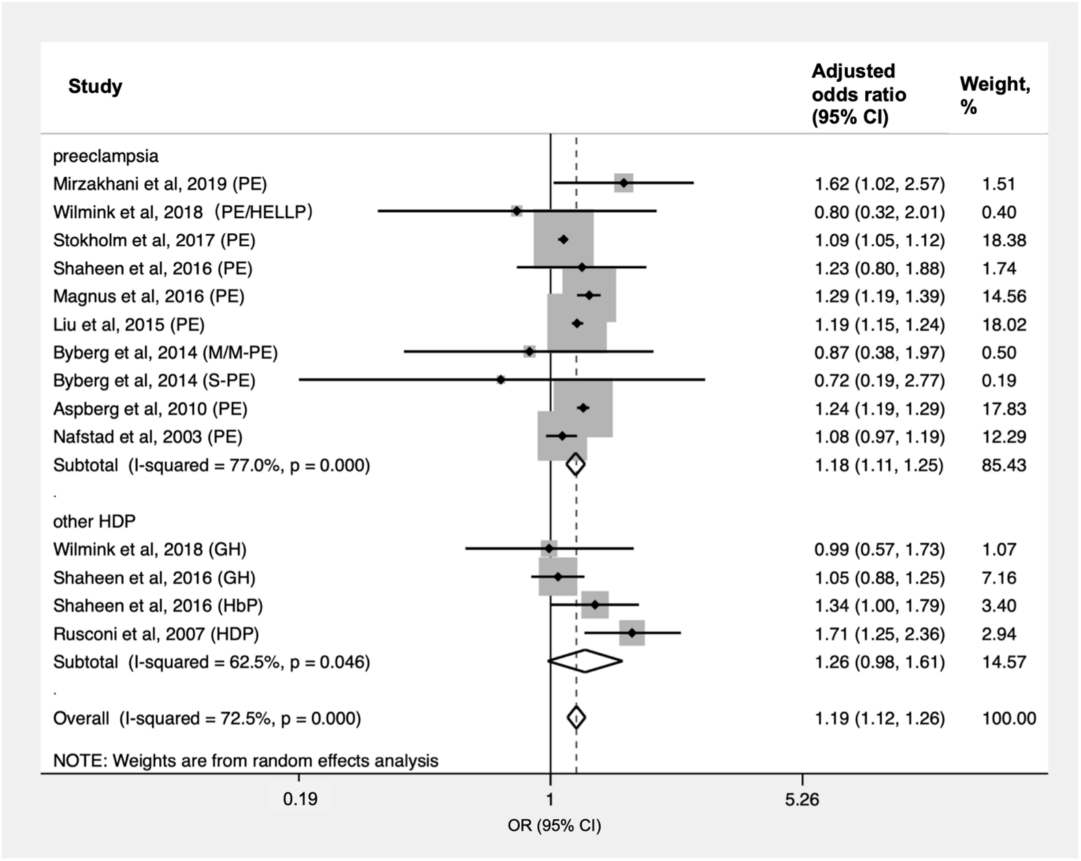
Supplemental eTable 4. Sensitivity analysis of the included studies.

Study omitted	Estimate	95% CI
Mirzakhani et al, 2019 (PE)	1.18	1.11 - 1.25
Wilmink et al, 2018 (GH)	1.19	1.12 - 1.26
Wilmink et al, 2018（PE/HELLP）	1.19	1.12 - 1.26
Stokholm et al, 2017 (PE)	1.21	1.15 - 1.27
Shaheen et al, 2016 (PE)	1.18	1.11 - 1.25
Shaheen et al, 2016 (GH)	1.20	1.12 - 1.26
Shaheen et al, 2016 (HbP)	1.18	1.11 - 1.23
Magnus et al, 2016 (PE)	1.16	1.09 - 1.23
Liu et al, 2015 (PE)	1.18	1.09 - 1.27
Byberg et al, 2014 (M/M-PE)	1.18	1.11 - 1.25
Byberg et al, 2014 (S-PE)	1.18	1.11 - 1.25
Aspberg et al, 2010 (PE)	1.17	1.10 - 1.24
Rusconi et al, 2007 (HDP)	1.17	1.11 - 1.24
Nafstad et al, 2003 (PE)	1.19	1.11 - 1.26

The estimate in each row indicates the pooled ORs after excluding that particular study.

Abbreviations: HDP, hypertension disorders in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; CI, confidence intervals.

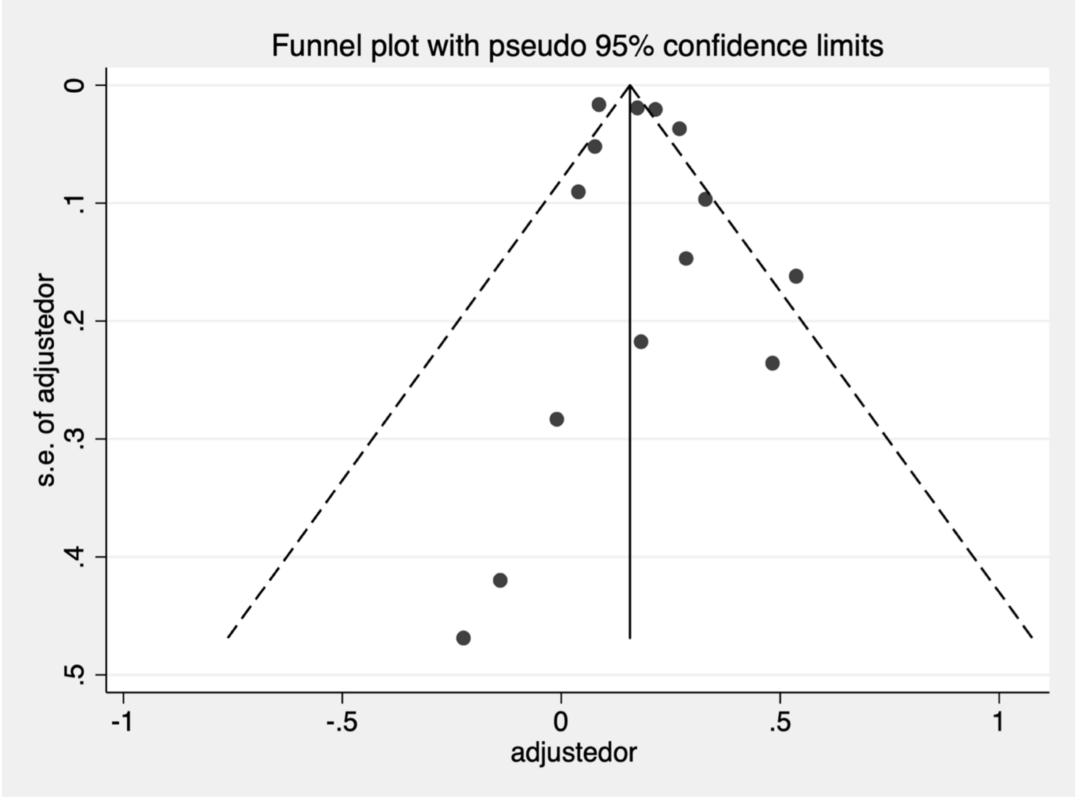
Supplemental eFigure 1. Forest plot of studies investigating the association between subtypes of HDP and asthma in offspring.



Forest plots displaying adjusted estimates examining the association between subtypes of HDP and asthma in offspring.

Abbreviations: PE, preeclampsia; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HDP, hypertension disorders in pregnancy; HbP, hypertension before pregnancy; M/M-PE, mild/moderate preeclampsia; S-PE, severe preeclampsia; HDP, hypertension disorders in pregnancy; OR, odds ratio; CI, confidence intervals.

Supplemental eFigure 2. Funnel plot of published studies investigating the association between HDP and asthma in offspring.



Abbreviations: ORs, odds ratios